Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

1- Theoretical part

1.1 Ethernet frame:

Below is a clear table summarizing the fields of an Ethernet frame :

Field	Size	Description
Preamble	7 bytes	Synchronizes the receiver with the sender using alternating bit pattern.
Start Frame Delimiter	1 byte	Marks the start of the frame.
Destination MAC Address	6 bytes	Identifies the receiving device.
Source MAC Address	6 bytes	Identifies the sending device.
Type/Length	2 bytes	Indicates the protocol type or payload length.
Data (Payload)	46–1500 bytes	Contains the actual transmitted data (with padding if needed).
Frame Check Sequence	4 bytes	Error detection using CRC to ensure data integrity.

1.2 ICMP Protocol :

Here is a concise table describing the main fields of an ICMP message:

Field	Size	Description
Type	1 byte	Indicates the ICMP message type (e.g., echo request, echo reply, error).
Code	1 byte	Provides more specific information about the message type.
Checksum	2 bytes	Used for error-checking of the ICMP message.
Rest of Header	4 bytes	Varies depending on the type/code (e.g., identifier, sequence number, etc.).
Data	Variable	Contains additional information (e.g., original packet data or payload).

Here's a simple example of an ICMP message, specifically an **Echo Request** (used by the `ping` command):

Field	Example Value	Description
Type	8	Echo Request message
Code	0	No additional code needed
Checksum	0x3A5F	Error-checking value (example)
Identifier	0x01	Identifies the ping session
Sequence Number	0x01	Sequence number of the request
Data	"Hello"	Payload sent to the destination

1.3 SMB Protocol:

SMB (Server Message Block) is a network file-sharing protocol that allows computers to **share files, folders, printers, and other resources** over a network.

Key Features

- **File Sharing:** Enables users to read, write, and manage files on remote systems.
- **Resource Sharing:** Provides access to printers and other shared devices.
- **Client–Server Model:** A client requests resources, and a server provides them.
- **Authentication:** Supports user login and permissions to control access.
- **Network Communication:** Typically runs over TCP/IP (commonly port 445).

Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

How It Works

1. A client connects to a server using SMB.
2. The user is authenticated (username/password).
3. The client requests access to shared resources (files, folders, printers).
4. The server responds and allows operations based on permissions.

Versions : SMB1 (**Old and insecure**), SMB2 (**Improved performance and efficiency**), SMB3 (**Adds security features like encryption**).

Here is a concise table summarizing the main **SMB2 header fields**:

Field	Size	Description
Protocol ID	4 bytes	Identifies the packet as SMB2 (e.g., `0xFE 'S' 'M' 'B'`).
Structure Size	2 bytes	Size of the SMB2 header (fixed value = 64 bytes).
Credit Charge	2 bytes	Number of credits consumed for the request (flow control).
Channel Sequence/Res.	2 bytes	Used for multi-channel communication or reserved.
Command	2 bytes	Specifies the SMB2 operation (e.g., read, write, negotiate).
Credit Request/Response	2 bytes	Number of credits requested or granted.
Flags	4 bytes	Indicates message properties (e.g., request/response, async).
Next Command	4 bytes	Offset to the next command (for compound requests).
Message ID	8 bytes	Unique identifier for matching requests and responses.
Process ID	4 bytes	Identifies the sending process.
Tree ID	4 bytes	Identifies the shared resource (e.g., a shared folder).
Session ID	8 bytes	Identifies the user session.
Signature	16 bytes	Used for message authentication and integrity (security).

In short: the SMB2 header contains control, identification, and security information needed to manage file-sharing operations efficiently.

Here is a simple example of an **SMB2 header** for a **READ request**:

Field	Example Value	Description
Protocol ID	0xFE534D42	Identifies SMB2 (`FE 'S' 'M' 'B'`)
Structure Size	64	Fixed size of SMB2 header
Credit Charge	1	One credit used for this request
Channel Sequence/Res.	0	Not using multi-channel
Command	0x0008	READ operation
Credit Request	32	Client requests 32 credits
Flags	0x00000000	Standard request (no special flags)
Next Command	0	No chained (compound) commands
Message ID	1001	Unique ID for this request
Process ID	0x1234	Client process identifier
Tree ID	0x0001	Refers to a shared folder
Session ID	0xABCDEF01	User session identifier
Signature	0000000000000000000	Not signed (for simplicity in this example)

Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

DCERPC (Distributed Computing Environment / Remote Procedure Call) is a protocol that allows programs to call procedures (functions) on a remote computer as if they were local, widely used in Windows and networked systems.

NetBIOS Session Service (NBT-SS) **provides a** session-layer channel **over TCP/IP for SMB communication.**

2- Practicales parts

Exercise 1

After configuring the ETHREAL software on each machine in the local network (as explained in the theoretical section), on each machine x with an IP address IPx, execute the following MS-DOS command: **ping IPy where** IPy is the IP address of another machine y. Then, stop the capture.

1. **Analyze the general information of the ETHERNET frame (information found in the frame line within section B of the ETHREAL window).**

Expanding the Ethernet section shows key fields:

• Destination MAC Address

- The physical address of the receiving machine
- Example: `00:1A:2B:3C:4D:5E`
- If pinging within the same LAN → this is the MAC of machine **y**

• Source MAC Address

- MAC address of the sending machine (machine **x**)

• Type (EtherType)

- Indicates the upper-layer protocol
- For **ping** traffic, you'll typically see:
  - `0x0800` → **IPv4**

2. **Retrieve different packet details (as explained in the theoretical section) from the ping command results, including:**

   - Time, source, destination, and protocol information from **section A** of the ETHREAL window.
   - Various headers displayed in **section B** for each protocol captured.

1. Information from Section A (Packet List Pane)

Section A shows a summary table of all captured packets. Each row corresponds to one packet.

Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

▯ Key Columns to Analyze

• Time

- Indicates when the packet was captured
- Measured in seconds since the start of capture
- Helps you observe:
  - Delay between request and reply
  - Packet timing (round-trip behavior)

☐ Example:

```
0.000000    → Echo Request
0.001200    → Echo Reply
```

• Source

- IP address of the sender
- For ping:
  - Request → machine **x**
  - Reply → machine **y**

• Destination

- IP address of the receiver
- Opposite of source:
  - Request → machine **y**
  - Reply → machine **x**

• Protocol

- Indicates the protocol used
- For ping:
  - Always **ICMP**

• Info (optional but useful)

- Displays message type:
  - "Echo (ping) request"
  - "Echo (ping) reply"

▯ What to conclude from Section A:

- You can track communication flow between machines
- You can measure response time (request → reply)
- You confirm that ICMP is used for ping

---

2. Information from Section B (Packet Details Pane)

Section B shows detailed protocol headers for the selected packet.

Each packet is structured like this:

`Frame → Ethernet → IP → ICMP`

A. Frame Header

- Frame number
- Frame size (bytes)
- Capture length

B. Ethernet Header

- **Source MAC address** → sender machine
- **Destination MAC address** → receiver machine
- **Type** → IPv4 (0x0800)

C. IP Header

Contains logical addressing and routing info:

- Source IP → machine x
- Destination IP → machine y
- TTL → packet lifetime
- Protocol → ICMP (value = 1)
- Total length → size of IP packet

D. ICMP Header (Ping-specific)

Most important for this experiment:

- Type:
  - 8 → Echo Request
  - 0 → Echo Reply
- Code → 0
- Identifier → same for request & reply
- Sequence number → increments with each ping
- Data → payload sent

Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

• Version : Usually IPv4 **(value = 4)**

• Header Length (IHL) :Typically 20 bytes **(no options)**

• Total Length:

- Size of the entire IP packet (header + ICMP)
- Usually around **60–84 bytes** in ping

• Protocol

- Indicates what is encapsulated inside IP:
  - `1` → ICMP (important for ping)

• TTL (Time To Live)

- Limits packet lifetime
- Typical values:
  - Windows: 128
  - Linux: 64

 It decreases by 1 at each router (not noticeable in same LAN)

• Identification / Flags / Fragmentation

- Used for fragmentation
- In a local ping:
  - Usually **no fragmentation**

ICMP Message Types

• Echo Request (Type 8)

- Sent by machine **x**

• Echo Reply (Type 0)

- Sent by machine **y**

 You will see alternating:

```
Echo (ping) request
Echo (ping) reply
```

Here are the main **ICMP header fields**:

---

1. Type (8 bits)

- Identifies the kind of ICMP message.
- Examples:
  - `8` → Echo Request (used by ping)
  - `0` → Echo Reply
  - `3` → Destination Unreachable
  - `11` → Time Exceeded

 This field tells *what the message is about*.

---

2. Code (8 bits)

- Provides more specific detail about the Type.
- Works as a subcategory of the Type field.

Examples:

- For Type 3 (Destination Unreachable):
  - `0` → Network unreachable
  - `1` → Host unreachable
  - `3` → Port unreachable

 Type + Code together fully describe the message.

---

3. Checksum (16 bits)

- Used for error-checking the ICMP message.
- Covers both header and data.
- Ensures data integrity during transmission.

 If checksum fails, the packet is discarded.

---

4. Rest of Header (32 bits)

- This part varies depending on the **Type and Code**.
- It's not fixed—different ICMP messages use it differently.

Examples:

- **Echo Request/Reply**:
  - Identifier (16 bits)
  - Sequence Number (16 bits)
- **Destination Unreachable**:

- Often unused (set to zero) or contains additional info

 This field is context-dependent.

---

5. Data (Variable length)

- Optional payload.
- Often contains:
  - Original IP header + first 8 bytes of the original packet (for error messages)
  - Arbitrary data (for ping)

 Used for diagnostics or matching requests/replies.

Example: ICMP Echo Request (Ping)

Field	Value Example
Type	8
Code	0
Checksum	Calculated
Identifier	1234
Sequence Number	1
Data	"hello"

3. In this lab, we focus on the network access layer header in the TCP/IP model, which corresponds to the physical and data link layers in the OSI model (ETHERNET header, section B). Fill in the following table for each protocol displayed.

Source @	Destination @	Protocol type
00:11:22:33:44:55	66:77:88:99:AA:BB	IPv4
AA:BB:CC:DD:EE:FF	FF:FF:FF:FF:FF:FF	ARP
12:34:56:78:9A:BC	98:76:54:32:10:FE	IPv6

Protocol (EtherType field)

This tells you what's inside the Ethernet frame:

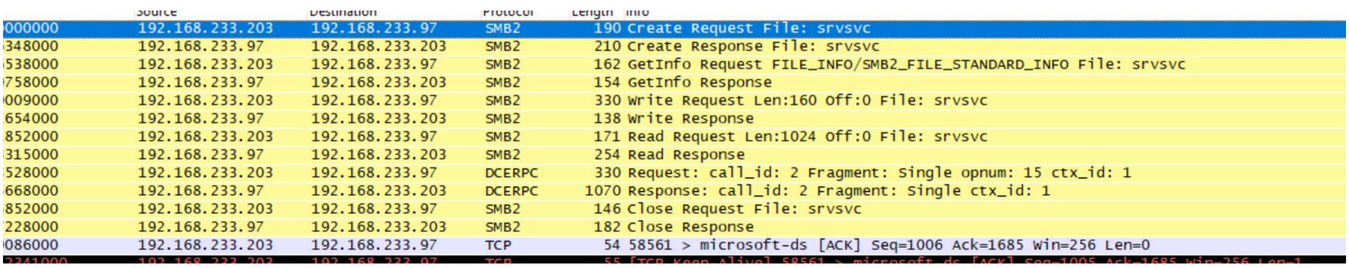
EtherType (Hex)	Protocol
`0x0800`	IPv4
`0x0806`	ARP
`0x86DD`	IPv6

Lab No. 4: Study of ETHERNET Frames (ICMP + SMB2)

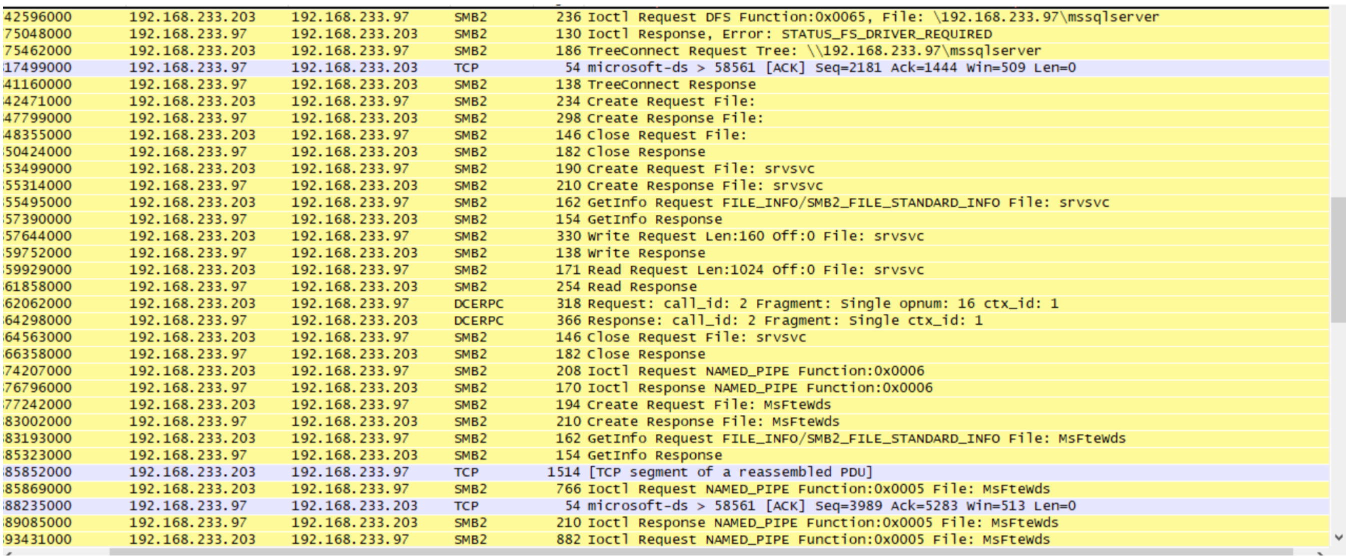
Exercise 2

In this exercise, we will determine the protocol that enables access to shared resources from a machine on the local network. (Create a network using a wifi access point and share a file on a server)

1. Launch ETHREAL, then access shared resources from another machine on the LAN.
   - For example, click Start > Run, type: **\\192.168.0.1**
   - Then, stop the capture and filter using : smb || smb2



	Source	Destination	Protocol	Length	Info
000000	192.168.233.203	192.168.233.97	SMB2	190	Create Request File: srvsvc
348000	192.168.233.97	192.168.233.203	SMB2	210	Create Response File: srvsvc
538000	192.168.233.203	192.168.233.97	SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc
758000	192.168.233.97	192.168.233.203	SMB2	154	GetInfo Response
009000	192.168.233.203	192.168.233.97	SMB2	330	Write Request Len:160 Off:0 File: srvsvc
654000	192.168.233.97	192.168.233.203	SMB2	138	Write Response
852000	192.168.233.203	192.168.233.97	SMB2	171	Read Request Len:1024 Off:0 File: srvsvc
315000	192.168.233.97	192.168.233.203	SMB2	254	Read Response
528000	192.168.233.203	192.168.233.97	DCERPC	330	Request: call_id: 2 Fragment: Single opnum: 15 ctx_id: 1
668000	192.168.233.97	192.168.233.203	DCERPC	1070	Response: call_id: 2 Fragment: Single ctx_id: 1
852000	192.168.233.203	192.168.233.97	SMB2	146	Close Request File: srvsvc
228000	192.168.233.97	192.168.233.203	SMB2	182	Close Response
086000	192.168.233.203	192.168.233.97	TCP	54	58561 > microsoft-ds [ACK] Seq=1006 Ack=1685 Win=256 Len=0



42596000	192.168.233.203	192.168.233.97	SMB2	236	Ioctl Request DFS Function:0x0065, File: \192.168.233.97\mssqlserver
75048000	192.168.233.97	192.168.233.203	SMB2	130	Ioctl Response, Error: STATUS_FS_DRIVER_REQUIRED
75462000	192.168.233.203	192.168.233.97	SMB2	186	TreeConnect Request Tree: \\192.168.233.97\mssqlserver
17499000	192.168.233.97	192.168.233.203	TCP	54	microsoft-ds > 58561 [ACK] Seq=2181 Ack=1444 Win=509 Len=0
41160000	192.168.233.97	192.168.233.203	SMB2	138	TreeConnect Response
42471000	192.168.233.203	192.168.233.97	SMB2	234	Create Request File:
47799000	192.168.233.97	192.168.233.203	SMB2	298	Create Response File:
48355000	192.168.233.203	192.168.233.97	SMB2	146	Close Request File:
50424000	192.168.233.97	192.168.233.203	SMB2	182	Close Response
53499000	192.168.233.203	192.168.233.97	SMB2	190	Create Request File: srvsvc
55314000	192.168.233.97	192.168.233.203	SMB2	210	Create Response File: srvsvc
55495000	192.168.233.203	192.168.233.97	SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: srvsvc
57390000	192.168.233.97	192.168.233.203	SMB2	154	GetInfo Response
57644000	192.168.233.203	192.168.233.97	SMB2	330	Write Request Len:160 Off:0 File: srvsvc
59752000	192.168.233.97	192.168.233.203	SMB2	138	Write Response
59929000	192.168.233.203	192.168.233.97	SMB2	171	Read Request Len:1024 Off:0 File: srvsvc
61858000	192.168.233.97	192.168.233.203	SMB2	254	Read Response
62062000	192.168.233.203	192.168.233.97	DCERPC	318	Request: call_id: 2 Fragment: Single opnum: 16 ctx_id: 1
64298000	192.168.233.97	192.168.233.203	DCERPC	366	Response: call_id: 2 Fragment: Single ctx_id: 1
64563000	192.168.233.203	192.168.233.97	SMB2	146	Close Request File: srvsvc
66358000	192.168.233.97	192.168.233.203	SMB2	182	Close Response
74207000	192.168.233.203	192.168.233.97	SMB2	208	Ioctl Request NAMED_PIPE Function:0x0006
76796000	192.168.233.97	192.168.233.203	SMB2	170	Ioctl Response NAMED_PIPE Function:0x0006
77242000	192.168.233.203	192.168.233.97	SMB2	194	Create Request File: MsFteWds
83002000	192.168.233.97	192.168.233.203	SMB2	210	Create Response File: MsFteWds
83193000	192.168.233.203	192.168.233.97	SMB2	162	GetInfo Request FILE_INFO/SMB2_FILE_STANDARD_INFO File: MsFteWds
85323000	192.168.233.97	192.168.233.203	SMB2	154	GetInfo Response
85852000	192.168.233.203	192.168.233.97	TCP	1514	[TCP segment of a reassembled PDU]
85869000	192.168.233.203	192.168.233.97	SMB2	766	Ioctl Request NAMED_PIPE Function:0x0005 File: MsFteWds
88235000	192.168.233.97	192.168.233.203	TCP	54	microsoft-ds > 58561 [ACK] Seq=3989 Ack=5283 Win=513 Len=0
89085000	192.168.233.97	192.168.233.203	SMB2	210	Ioctl Response NAMED_PIPE Function:0x0005 File: MsFteWds
93431000	192.168.233.203	192.168.233.97	SMB2	882	Ioctl Request NAMED_PIPE Function:0x0005 File: MsFteWds

2. Identify the protocol responsible for this activity.

   SMB2

3. Complete the previous table.

Protocol	Source Address	Destination Address	Protocol Type
DCERPC	00:1A:2B:3C:4D:5E	00:5E:4D:3C:2B:1B	0x0800 (IPv4)
SMB2	00:1A:2B:3C:4D:5E	00:5E:4D:3C:2B:1B	0x0800 (IPv4)

4. Identify any differences (if applicable) between the resulting table and the previous ones.

Summary:

- At the **Ethernet layer**, both ICMP and SMB frames share the same basic structure (MAC addresses, Ethertype, FCS).
- The **differences appear in the payload and protocol fields**:
  - ICMP frames: `IP Protocol = 1`, no TCP/UDP, small payload.
  - SMB frames: `IP Protocol = 6`, TCP header present, larger and variable payload.